Numerical Methods — Laboratory Activity 02

Fitting a Curve to the Dam: Levenberg–Marquardt, Residuals, and Area Under the Level

Model: double logistic, h(t) = c + a■/(1 + exp(-k■(t – t■■))) + a■/(1 + exp(-k■(t – t■■))). Initial guess p■ = [14, 8, 0.5, 31, –4, 0.3, 38].

Data audit: the activity sheet says 96 readings in one day, while the supplied workbook contains 288 15-minute readings spanning 72 hours. All supplied workbook readings were used unchanged.

Parameter	Estimate	SE	t	p-value
c	14.3493	0.0097	1483.028	<1×10■¹■
a■	8.9157	0.7532	11.837	<1×10■¹■
k■	0.5117	0.0213	24.048	<1×10■¹■
t■■	31.5198	0.0906	348.068	<1×10■¹■
a■	-3.7210	0.7595	-4.899	1.63e-06
k■	0.2831	0.0138	20.589	<1×10■¹■
t■■	37.5466	1.0575	35.506	<1×10■¹■

SSE (m²)	1.801659	**SST (m²)**	2034.077941
R^2	0.999114	**s (m)**	0.080072
n, p, df	288, 7, 281	**Max |residual| (m)**	0.2208
Max dh/dt (m/h)	1.0060	**Time of max rate**	2026-07-22 07:20
quad area (m·h)	1260.970821	**quad abs. error**	5.090e-07
Raw trapezoid (m·h)	1260.996250	**quad – trapezoid**	-0.025429

Residual reading: residuals are close to zero in average, but there are long same-sign runs; therefore the high R^2 should not be treated as proof of a perfect shape match. The largest residual is 0.2208 m, exceeding the logger's 0.01 m resolution by 0.2108 m.

Second derivative reading: the main filling episode accelerates before the maximum dh/dt and decelerates after it, consistent with a rise toward a peak filling rate followed by settling/recession.